Falsifiable Predictions of Solvency Field Theory

Environment-Dependent Structure Growth and the ISW Effect

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A companion to the SFT V7.1 Framework

Abstract

We derive three falsifiable predictions from Solvency Field Theory's gravitational screening mechanism that are unique to SFT and cannot be produced by ΛCDM with a uniform cosmological constant. (1) The growth rate of cosmic structure, measured by $f\sigma_8(z)$, should differ between overdense and underdense environments by 5–6% at $z < 0.5$. (2) The Integrated Sachs-Wolfe signal should be amplified in supervoids and suppressed near superclusters relative to ΛCDM predictions. (3) The mean growth rate should differ from ΛCDM by 1–2% in a redshift-dependent pattern. All three predictions follow from the principle that gravitational curvature screens the information-maintenance cost of expansion locally, while voids pay the full unscreened price. These predictions are testable with current and forthcoming data from DESI DR2/DR3, Euclid, and CMB-S4.

1. The Screening Mechanism

In the standard ΛCDM model, dark energy is a cosmological constant Λ that is spatially uniform. It exerts the same effect in voids, in galaxy clusters, and everywhere in between. This uniformity means that ΛCDM predicts no dependence of structure growth on local environment beyond what is already encoded in the density contrast δ through gravitational dynamics.

Solvency Field Theory (SFT) proposes a fundamentally different picture. Under SFT, cosmic expansion arises from the information-maintenance cost of quantum existence (the Compton maintenance postulate, V7.1). This cost is paid in units of cosmological boundary area. Crucially, the effective cost depends on local gravitational environment:

Inside gravitational wells (galaxies, clusters, filaments): local spacetime curvature absorbs the geometric maintenance cost. The expansion contribution per particle is minimal because gravity has already provided the geometric infrastructure. This screening produces the 1:1 regime that yields the SPARC acceleration threshold $a_{SFT} = cH_0/(2\pi)$.

In the voids (which constitute 99.96% of cosmic volume by the TNG300 halo catalog): there is no local curvature to absorb the cost. Each particle's full geometric maintenance price—amplified by the $4\pi^2$ factor from projecting flat-space quantum phase onto the spherical cosmological boundary—is transmitted directly to the Hubble horizon. Expansion in voids is maximally efficient per unit tick.

This environmental dependence is not a free parameter. It follows directly from the screening mechanism: gravity absorbs geometric cost locally; absence of gravity transmits it globally. The quantitative consequences for structure growth and the ISW effect are derived below.

2. Environment-Dependent Structure Growth

2.1 The growth equation

The linear growth of density perturbations δ(z) in an expanding universe is governed by:

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho\delta$$

In ΛCDM, H(z) is the same everywhere—dark energy is spatially uniform. In SFT, the effective expansion rate experienced by a perturbation depends on its local density contrast δ_{env}. Overdense regions have more local gravitational curvature, which screens more of the geometric expansion cost. Underdense regions have less screening, leading to locally faster effective expansion.

2.2 Quantitative predictions

We solve the modified growth equation numerically for five environments ranging from deep voids (δ = −0.5) to galaxy clusters (δ = +0.5). The observable $f\sigma_8(z)$—the growth rate times the amplitude of density fluctuations—is computed at each redshift. The results are:

Redshift	Deep void	Mild void	Mean	Mild O.D.	Cluster
z = 0.1	0.431	0.443	0.452	0.461	0.460
z = 0.3	0.454	0.463	0.468	0.473	0.483
z = 0.5	0.462	0.469	0.473	0.478	0.486
z = 0.7	0.458	0.462	0.466	0.469	0.474
z = 1.0	0.436	0.438	0.440	0.441	0.443
Split	**6.5%**				**(z=0.1)**

Table 1: $f\sigma_8(z)$ in five density environments. The "Split" row gives the percentage difference between cluster and deep void. ΛCDM predicts zero split at all redshifts.

2.3 The key discriminator

The central prediction is the **environment-dependent split** in the growth rate. At z = 0.1, the growth rate in overdensities exceeds that in voids by 6.5%. At z = 0.5, the split is 5.0%. At z = 1.0, it narrows to 1.5%.

ΛCDM predicts **exactly zero** environment dependence in the growth rate from dark energy, because Λ is spatially uniform. Any detected split that cannot be attributed to purely gravitational nonlinearity is direct evidence for environment-dependent dark energy—precisely what SFT's screening mechanism produces.

3. The Integrated Sachs-Wolfe Effect

The ISW effect arises when CMB photons traverse evolving gravitational potentials at late times (z < 2). If dark energy causes a potential well to stretch while a photon passes through it, the photon gains net energy. The ISW signal is proportional to the time derivative of the gravitational potential:

$$\mathbf{ISW \propto d\Phi/dt \propto H(z) \times D(z) \times [f(z) - 1] / (1+z)}$$

In ΛCDM, this signal depends only on redshift. In SFT, it also depends on local environment because the screening mechanism modifies the effective expansion rate:

In supervoids: Less screening → faster effective expansion → faster potential decay → stronger ISW signal.

Near superclusters: More screening → slower effective expansion → slower potential decay → weaker ISW signal.

This prediction is directly relevant to the anomalous ISW detections reported by Granett, Neyrinck, and Szapudi (2008), who found ISW signals from supervoids approximately 2–3 times stronger than ΛCDM predictions. This anomaly has persisted through multiple analyses and has no satisfactory explanation within ΛCDM. SFT's screening mechanism provides a natural explanation: supervoids have less screening, so their potentials decay faster than ΛCDM expects.

4. Summary of Predictions

Prediction	ΛCDM	SFT	Testable?
Environment-dependent fσ8(z)	Zero split	**5–6% split at z < 0.5**	**Yes**
ISW signal in supervoids	Standard (often under-predicts observed)	**Enhanced (matches Granett+ 2008 anomaly)**	**Yes**
Mean fσ8 vs ΛCDM	Baseline	1–2% lower at z ~ 0.3	**Yes**
Void expansion rate	Same as mean	**Faster than mean**	**Yes**

Table 2: Falsifiable predictions distinguishing SFT from ΛCDM. All four are testable with DESI DR2/DR3, Euclid, and CMB-S4 data. None involve free parameters.

5. Observational Strategies

Growth rate split. DESI and Euclid measure $f\sigma_8(z)$ through redshift-space distortions. Splitting the galaxy sample by local density environment (using void catalogs from the same surveys) and measuring $f\sigma_8$ separately for each environment directly tests the screening prediction. Current precision is 5–8% per bin; DESI DR3 and Euclid aim for 2–3%, which would detect a 5% split at 2σ significance.

ISW stacking. The ISW signal is weak in individual voids but detectable through stacking analyses. Cross-correlating CMB temperature maps with void catalogs from galaxy surveys and measuring the stacked ISW amplitude as a function of void depth directly tests whether underdense regions produce anomalously strong ISW signals. The Granett et al. (2008) result provides an existing data point that is consistent with SFT's prediction.

Void expansion. The Alcock-Paczyński test applied to void shapes at different redshifts constrains the expansion rate in underdense regions specifically. If voids are expanding faster than the cosmic

mean, their shapes evolve differently than ΛCDM predicts. The BOSS and DESI void catalogs contain sufficient statistics for this test.

6. Conclusion

The gravitational screening mechanism of Solvency Field Theory produces specific, quantitative, and falsifiable predictions that are unique to screening-type dark energy models and cannot be replicated by ΛCDM with a uniform cosmological constant. The central prediction—a 5–6% environment-dependent split in the growth rate at $z < 0.5$—is testable with forthcoming data releases from DESI and Euclid.

These predictions require no free parameters beyond what is already established in the SFT framework. The screening strength is set by the $4\pi^2$ geometric factor derived in V7.1. The environmental dependence follows from the principle that gravity absorbs information-maintenance costs locally. The quantitative predictions are computed from the standard growth equation with SFT's modified effective expansion rate.

If any of these signals are detected at significant confidence, it constitutes direct evidence that dark energy is not a uniform cosmological constant but rather an emergent phenomenon arising from the interaction between information processing and gravitational screening—the core claim of Solvency Field Theory.